PRACTICAL LABORATORY ASSIGNMENT

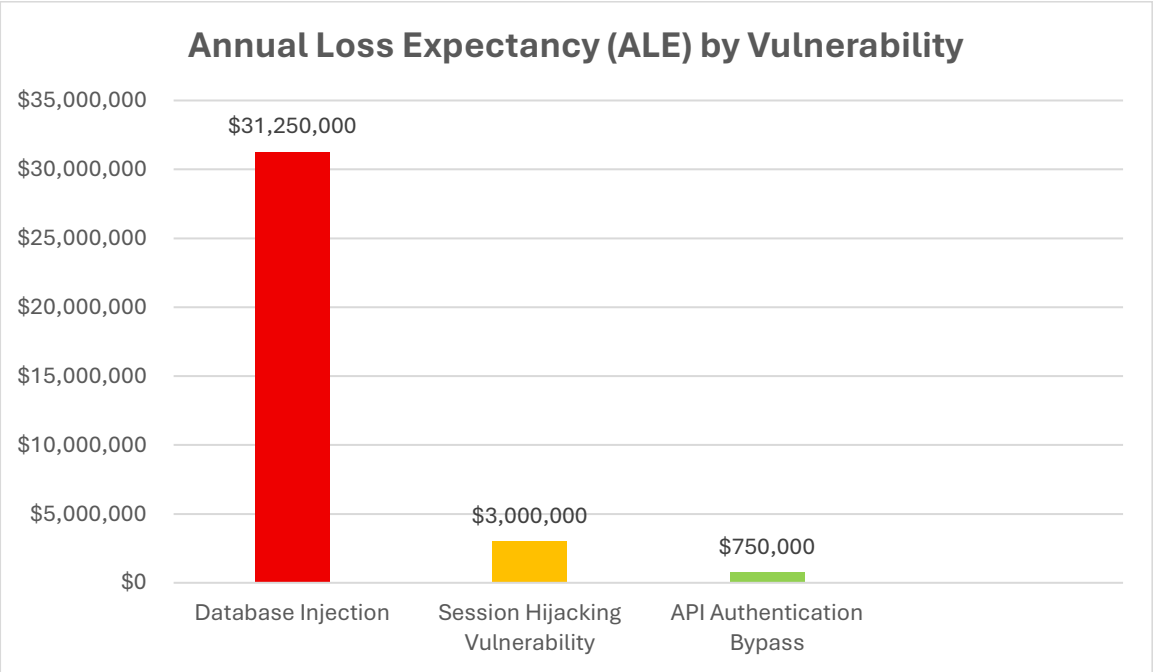
Full Name	OSAWARU PRAISE
Registration Number	C11/26/CGRCE/17564
Date Submitted	1st November 2026
Course	GRC 103 Week 2 Lab B

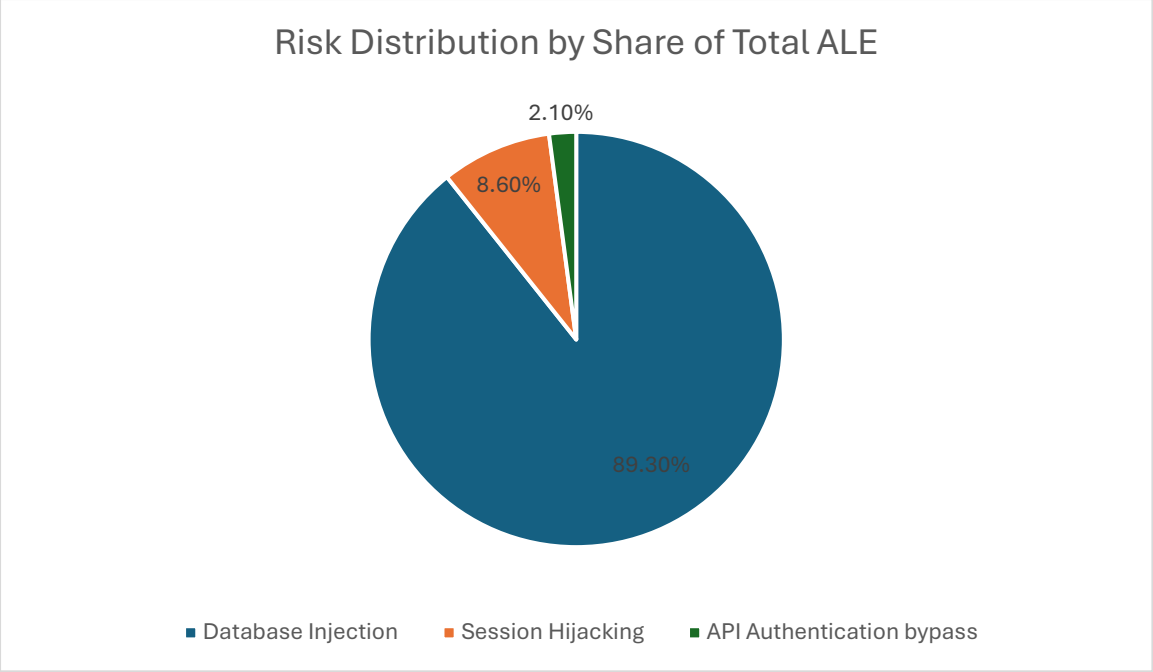
Phase 1: Risk Exposure Calculations

Task 1 & 2: Calculate Annualized Loss Expectancy (ALE) and Prioritize Risks

Vulnerability	**SLE**	**ARO**	**ALE**	**Priority**
DB Injection	500,000 × $250 = $125,000,000	0.25	$31,250,000	Critical
Session Hijacking	5,000 × $1,500 = $7,500,000	0.40	$3,000,000	High
API Auth Bypass	$5,000,000	0.15	$750,000	Medium
Total			$35,000,000	

Task 3: Create Risk Visualization





Risk Heat Map

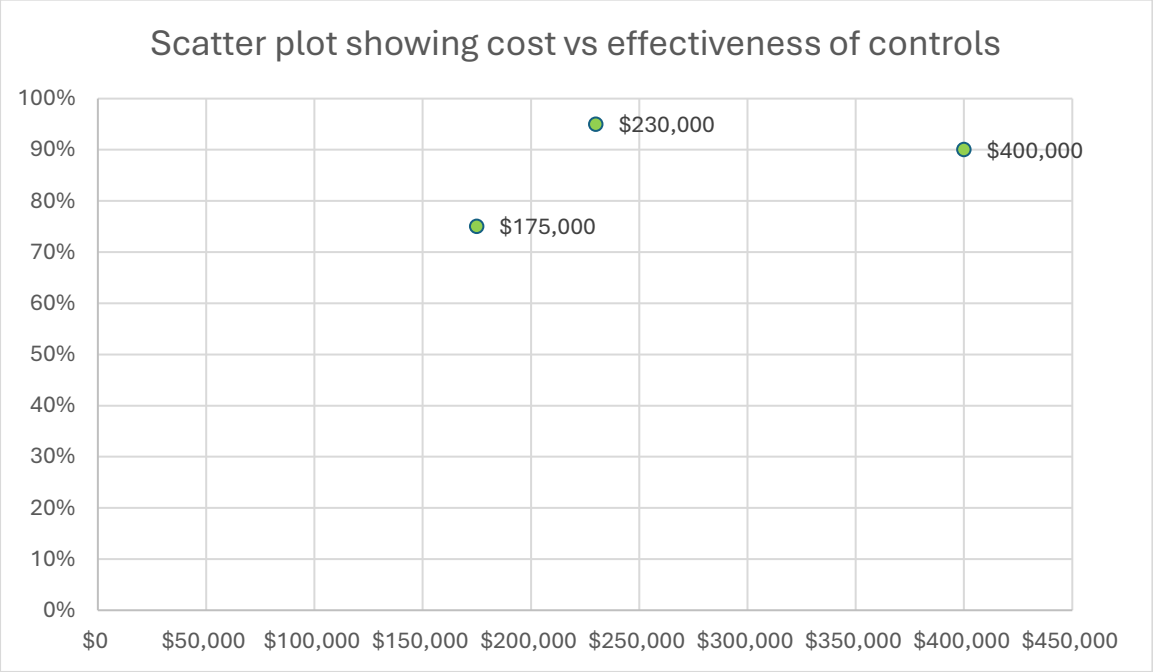
High (>30%)	Medium	High Session hijacking 40% | $7.5M SLE	High
Medium (20-30%)	Low	Medium	High Database Injection 25% | $125M SLE
Low (<20%)	Low	Low API auth bypass 15% | $5.0M SLE	Medium

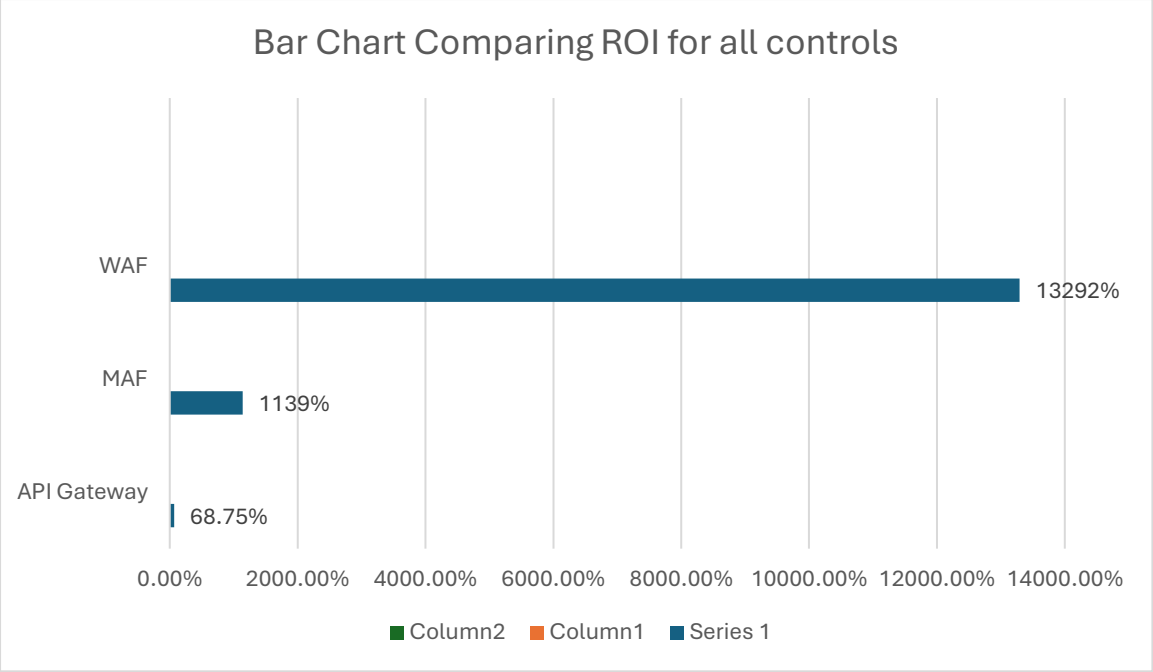
Phase 2: Control Evaluation

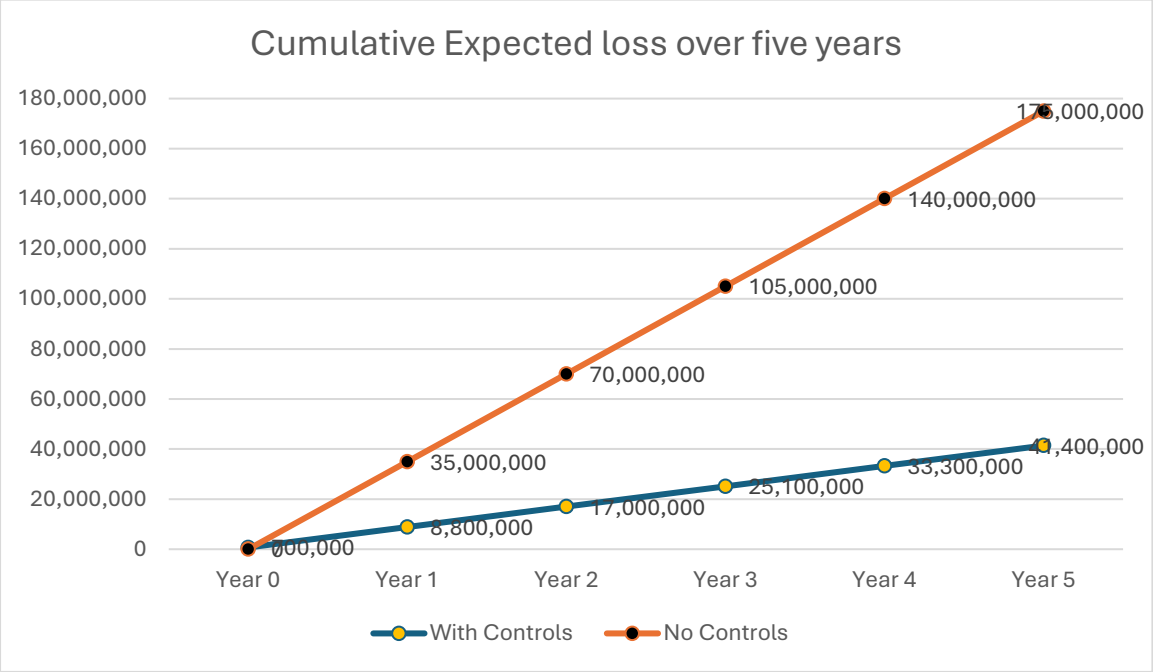
Task 4: Cost-Benefit Analysis

Control	Initial	Annual maintenance	Risk reduction/yr	Year-1 ROI	Payback
API Gateway (90%)	$350,000	$50,000	$675,000	68.75%	~6.7 months
WAF (75%)	$150,000	$25,000	$23,437,500	13,292.9%	~2.3 days
MFA (95%)	$200,000	$30,000	$2,850,000	1,139.1%	~26 days

Task 5: Control Selection Analysis




Bar Chart Comparing ROI for all controls
WAF
MAF
API Gateway
13292%
1139%
68.75%
0.00%
2000.00%
4000.00%
6000.00%
8000.00%
10000.00%
12000.00%
14000.00%
Column2
Column1
Series 1


Cumulative Expected loss over five years
180,000,000
160,000,000
140,000,000
120,000,000
100,000,000
80,000,000
60,000,000
40,000,000
20,000,000
0
140,000,000
105,000,000
70,000,000
35,000,000
33,300,000
25,100,000
17,000,000
8,800,000
Year 0
Year 1
Year 2
Year 3
Year 4
Year 5
With Controls
No Controls

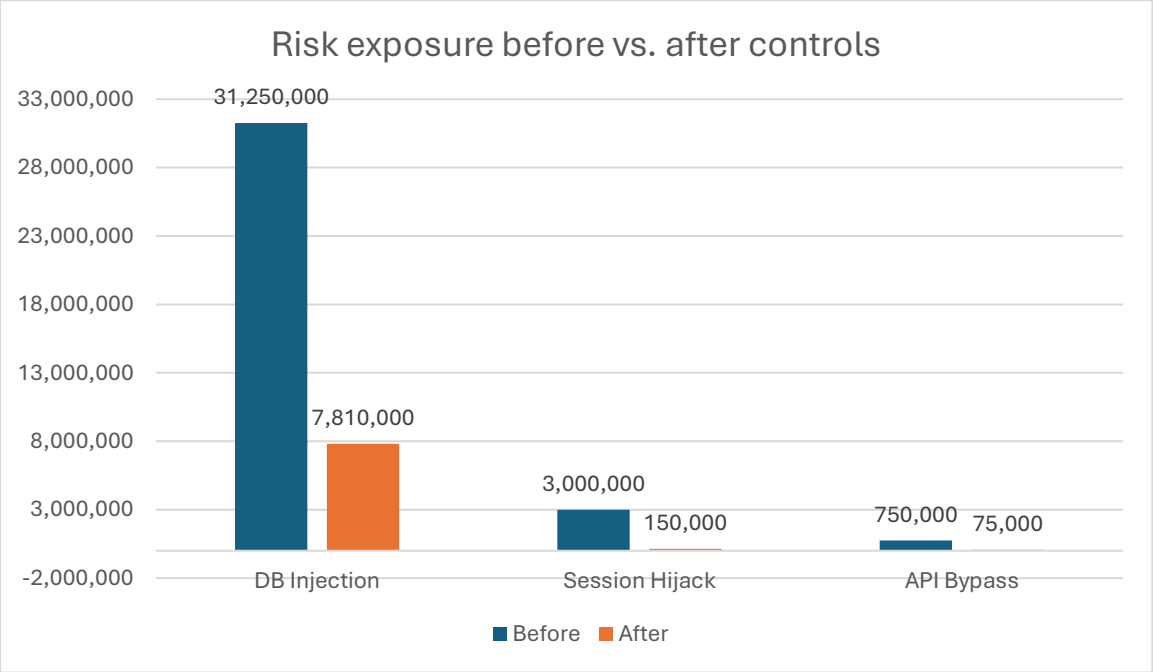
Phase 3: Executive Reporting

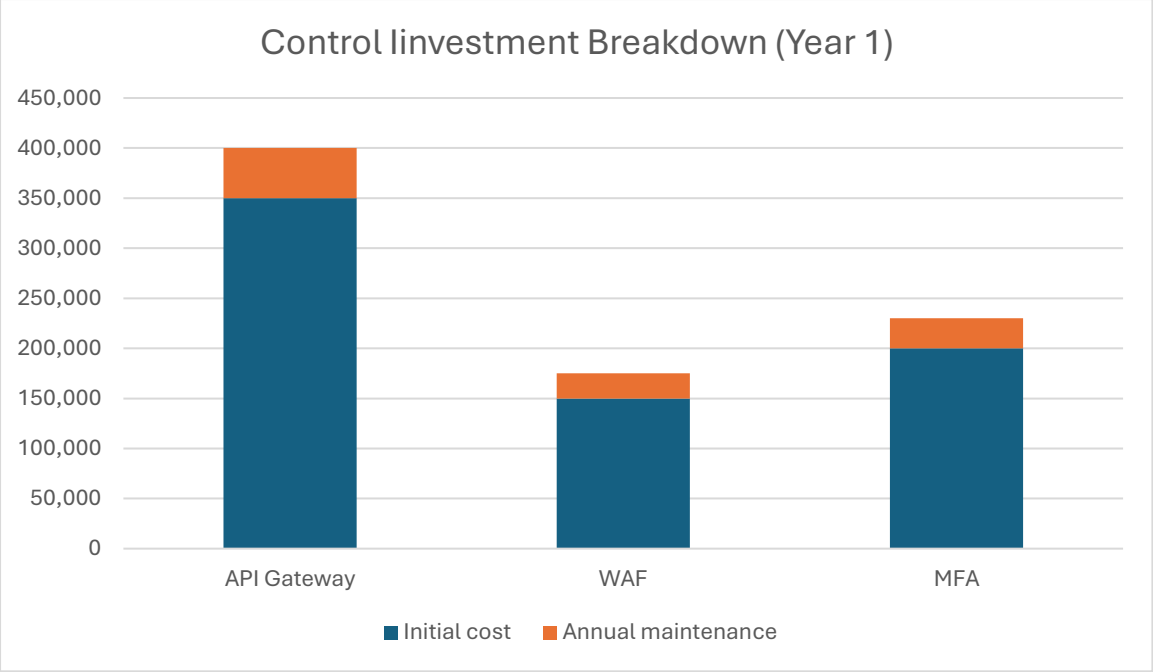
Task 6: Creative Executive Dashboard

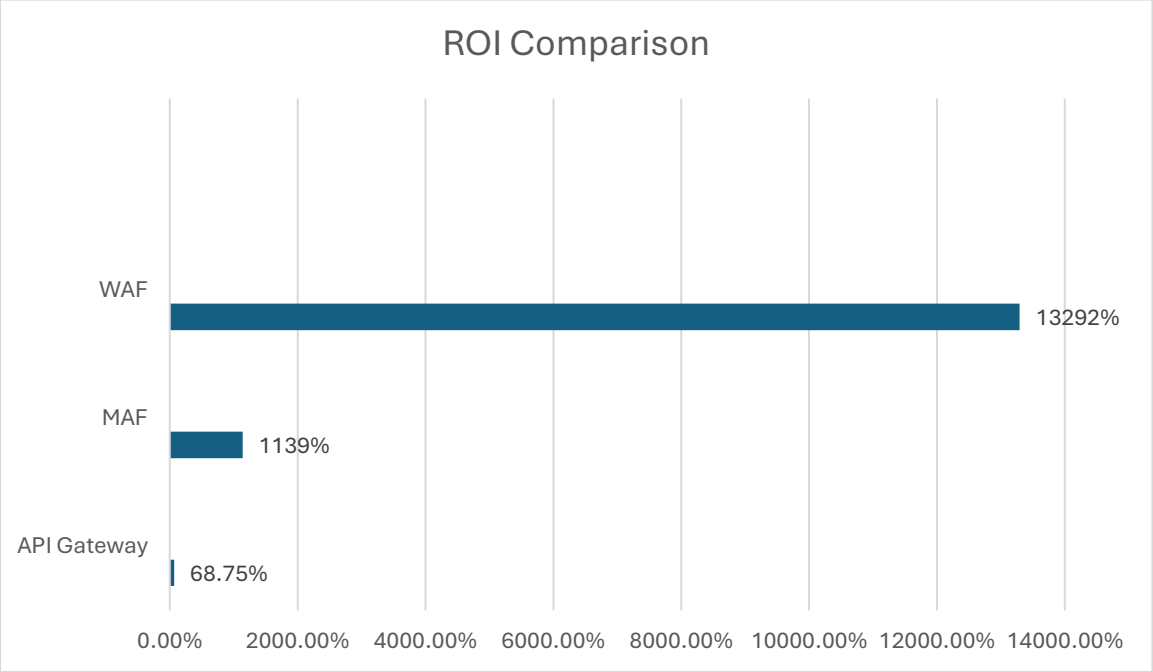
Top 3 vulnerabilities, recommended controls, expected risk reduction and ROI

Total ALE today	ALE after controls	Risk reduction	Year-1 investment	Combined ROI
$35.0M	$8.04M	$26.96M (77%)	$805K	3,249%

Top risk	ALE	Recommended control	Year-1 cost	Risk reduced	ROI
1. Database injection	$31.25M	WAF (75%)	$175K	$23.44M	13,293%
2. Session hijacking	$3.00M	MFA (95%)	$230K	$2.85M	1,139%
3. API auth bypass	$0.75M	API Gateway (90%)	$400K	$0.68M	69%
Total	$35.0M		$805K	$26.96M	3,249%




Control Iinvestment Breakdown (Year 1)
450,000
400,000
350,000
300,000
250,000
200,000
150,000
100,000
50,000
0
API Gateway
WAF
MFA
Initial cost
Annual maintenance


ROI Comparison
WAF
13292%
MAF
1139%
API Gateway
68.75%
0.00%
2000.00%
4000.00%
6000.00%
8000.00%
10000.00%
12000.00%
14000.00%

Task 7: Risk Treatmnet Timeline

Horizon	Action	Days	Type
Immediate (first 30 days)	Deploy WAF in blocking mode	0–30	Costed control
	Harden sessions (shorter timeouts, token rotation)	0–14	Supporting
	Alerting for injection and hijack attempts	7–30	Supporting
Short-term (by day 90)	Roll out MFA (pilot, then all users)	15–90	Costed control
	Deploy API Gateway	30–90	Costed control
	Penetration test to validate controls	75–90	Supporting
Long-term (by day 365)	Secure code fixes (parameterized queries, input validation)	90–270	Supporting
	Staff training and incident response drills	120–300	Supporting
	Continuous monitoring, quarterly ALE review	90–365	Supporting